MCO-1D PROGRAMME

The Laminar Separation Bubble

and why the learning system cannot currently see it

Proposed hypothesis H-LSB-1

September 2026

Summary

At the Reynolds numbers the MCO-1D flies at, the upper-surface boundary layer separates, transitions, and usually reattaches — enclosing a bubble of recirculating flow. Sometimes it fails to reattach. That failure is sudden, local to one wing, and costs lift without warning.

All 162 existing hypotheses score mean values. A bubble that bursts one stroke in twenty barely moves a mean. The genome would converge on a configuration that works well on average and occasionally drops a wing, and nothing in the system would object.

H-LSB-1 scores **stroke-to-stroke consistency** instead, and uses the eight independent current sensors to separate bubble bursting from gusts. It is the one significant gap the physics audit identified in the hypothesis registry.

1. What a laminar separation bubble is

Air arriving at the leading edge forms a boundary layer that begins laminar — smooth, ordered, layered flow. Laminar layers carry very little momentum near the wall, which makes them fragile.

Past the suction peak on the upper surface, pressure rises with distance downstream. This adverse gradient pushes against the flow. A laminar layer has too little near-wall momentum to fight it and lifts off the surface — laminar separation.

The separated shear layer is unstable. Within a short distance it transitions to turbulence, and turbulent flow mixes high-momentum air from the free stream down toward the wall. That re-energised layer can usually overcome the gradient and reattaches to the surface. Between the separation point and the reattachment point sits a region of slow recirculating flow: the bubble.

The two states

Bubble present	Separation, transition, and reattachment all occur. The bubble is a thin closed region, typically 10–30% of chord long. Lift is close to the attached-flow value. Drag is elevated because the bubble displaces the effective body shape and thickens the downstream boundary layer. This is the normal condition.
Bubble burst	Transition happens too late, or the gradient is too severe, and the shear layer never reattaches. Flow separates from the bubble location all the way to the trailing edge. Lift drops sharply, drag rises sharply. This is a stall — but it arrives without the AoA warning signs a conventional trailing-edge stall gives.

> **Why bursting is the dangerous part**
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> A bubble that sits stably is a known, bounded drag penalty. You can design around it. Bursting is different because it is **intermittent and condition-sensitive**. Small changes in Reynolds number, angle of attack, surface roughness, or free-stream turbulence flip the outcome. Two consecutive strokes at nominally identical commanded conditions can behave differently — one attached, one separated.
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> That nondeterminism is what makes it invisible to a system that measures averages.

2. Why the MCO-1D is squarely in the affected regime

Bubble formation is a low-Reynolds-number phenomenon. Above roughly Re 500,000 natural transition happens early enough that the layer is turbulent before it reaches the adverse gradient, and no bubble forms. Below roughly Re 30,000 the flow separates so readily that it never reattaches at all. The bubble regime sits between — and the MCO-1D sits in the middle of it, at every flight condition.

Flight condition	V (m/s)	Re (165 mm chord)	Bubble behaviour
Slow SAR orbit	3.0	33,000	Lower edge — bubbles marginal
Best glide	5.2	57,200	Prime bubble regime
Wingtip, hover	8.73	96,000	Prime bubble regime
Cruise	9.0	99,000	Prime bubble regime
Stoop / dive	20.0	220,000	Upper edge — bubbles shorten

Reynolds number computed on the 165 mm root chord at sea level, $\nu = 1.5 \times 10^{-5}$ m^2/s.

The S1223 profile intensifies the problem. It achieves its high lift coefficient through aggressive camber, which produces a strong suction peak near the leading edge and therefore a steep adverse gradient behind it. The same feature that makes it a good high-lift section makes it prone to bubbling.

Flapping adds a dimension that fixed-wing analysis does not have to consider. Over a single stroke the local velocity at any spanwise station sweeps from near zero at reversal to peak at mid-stroke. Local Reynolds number therefore sweeps across the entire bubble-sensitive range twice per cycle, at 1.87 Hz. The bubble is not a static feature to be designed around — it forms, migrates, and may burst within every stroke.

3. The measurement blind spot

Every hypothesis currently in the registry scores a mean. Mean efficiency, mean descent rate, mean pointing error, mean regenerative power. This is the right approach for almost everything the aircraft does — and it is precisely wrong for bubble bursting.

A worked comparison

Consider two candidate wing configurations evaluated over 100 strokes. Configuration A never bursts. Configuration B bursts on 5% of strokes, losing roughly 40% of lift on each affected stroke.

Metric over 100 strokes	Config A	Config B	Separation
Strokes with attached flow	100	95	—
Strokes with burst	0	5	—
Mean lift (normalised)	1.000	0.980	**2.0% — within noise**
Mean current draw	1.000	0.988	**1.2% — within noise**
Stroke-to-stroke variance	0.004	0.031	**7.8× — unmistakable**

On mean lift the two configurations are 2% apart. Flight-to-flight variation from wind, battery state, temperature and payload comfortably exceeds 2%, so the genome cannot reliably distinguish them — and B may even score better on a given flight by chance.

On variance they are nearly eight times apart. That is not a marginal signal requiring careful statistics. It is obvious in a single evaluation window.

> **The consequence if this is left unaddressed**
>
> The genome optimises what it is scored on. With only mean-value metrics available, it will happily converge on Configuration B — a configuration that performs well on average and sheds a wing unpredictably once every twenty strokes.
>
> On a SAR aircraft flying low over difficult terrain, an intermittent single-wing lift loss during an approach is a meaningfully worse outcome than a few percent of extra cruise drag. The system currently has no way to express that preference.

4. Distinguishing a bubble burst from a gust

Elevated variance alone is not proof. A gusty day also produces stroke-to-stroke variation in current draw. If H-LSB-1 cannot separate the two it will generate false positives on windy days and be ignored — the fate of any alert that cries wolf.

The separation comes from spatial correlation across the eight wings, and it is clean because the two phenomena have fundamentally different structure.

Gust	A parcel of moving air with physical extent, typically metres. It reaches the wing array as a coherent front and propagates through the metachronal cascade in sequence. Adjacent wings see correlated disturbances with a consistent time lag set by the cascade phase. Cross-wing correlation is **high and lagged**.
Bubble burst	A boundary-layer instability local to one wing surface. It depends on that wing's own instantaneous local Re, AoA, and surface state. Wing 3 may burst while wing 4, 20 mm away and 28° out of phase, does not. Cross-wing correlation is **near zero**.

The test is therefore a two-part conjunction: elevated per-wing variance, combined with low cross-wing correlation. Elevated variance with high correlation is turbulence and belongs to H-GS-1. Elevated variance with low correlation is a boundary-layer event and belongs to H-LSB-1.

This test requires the per-wing architecture

Eight independent motors, eight independent ODrive controllers, eight independent current sensors sampling at 50 Hz. A paired-motor or four-motor system measures the sum across two or more wings and cannot compute a cross-wing correlation at all.

The individual-wing architecture chosen for control authority turns out to enable a diagnostic capability that was not part of its original justification.

5. Hypothesis specification

Identifier	H-LSB-1
Label	Laminar separation bubble stability — find wing configurations that maintain consistent attached flow across the stroke cycle
Target axis	Stability
Genome axis	_MW_STABILITY, with secondary reinforcement to _MW_PITCH
Evaluation phases	Cruise, glide — steady conditions where the variance measurement is meaningful
Lever zones	CRUISE (55–68%), GLIDE_SOAR (28–38%)
Priority	1 — normal discovery rotation, after envelope characterisation
Envelope gate	None. Applies in forward flight regardless of hover capability
Manoeuvre only	No

Scoring

The joint score weights consistency far above efficiency. A slightly draggier configuration that never bursts is worth more on this aircraft than an efficient one that occasionally does.

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score = 0.70 × (1 − normalised_variance) + 0.30 × (1 − normalised_mean_drag)
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New SG task	Definition
lsb_stroke_variance	Stroke-to-stroke variance in per-wing current at matched lever and airspeed, averaged across all eight wings. Lower is better. This is the primary term.
lsb_cross_correlation	Mean pairwise correlation of the per-wing current residuals across the eight channels. Used as a gate, not scored: high correlation disqualifies the sample as a gust event.
lsb_burst_rate	Fraction of strokes in the evaluation window showing a current drop beyond threshold with low cross-correlation. Reported in the debrief; diagnostic rather than scored.

Parameter space explored

Variable	Range	Why it matters
Angle of attack	8° – 24°	Primary driver of adverse gradient strength. Bubble location and burst threshold both move with AoA.
Stroke frequency	1.4 – 2.6 Hz	Sets the local Reynolds sweep range over the stroke, and the rate at which conditions change.
Stroke amplitude	84° – 108°	Changes the velocity profile along the span and the duration spent at low local Re near reversal.
Cascade phase	fixed at 28°	Held constant so that inter-wing wake effects do not confound the correlation measurement.

The search objective is a stability basin, not a peak. A configuration that holds low variance across a range of AoA is more valuable than one that achieves the lowest variance at a single point, because real flight does not hold a single point.

6. Integration with existing systems

Wear modality	Already tracks mid-stroke current peak timing per wing to detect winglet stiffness change. An LSB burst also perturbs peak timing. The wear baseline must exclude strokes flagged as burst events, or aerodynamic intermittency will be misread as mechanical wear.

Anomaly archaeology	Would likely detect the burst signature eventually as an unexplained variance cluster. Naming the phenomenon lets the genome search against it deliberately rather than stumbling onto it, and gives the finding a physical interpretation.
H-GS-1 gust suppression	Consumes the same per-wing current anomaly stream. The cross-correlation gate is what keeps the two from fighting over the same events. High correlation routes to H-GS-1; low correlation routes to H-LSB-1.
ALC alerting	A sustained rise in burst rate is a tier-1 cruise observation — informational, not urgent. A burst rate above roughly 15% during an approach warrants tier 2, because that is where an intermittent lift loss has the least altitude to recover in.
Turbulator testing	The variance metric is the correct instrument for evaluating boundary-layer trip tape. Fly the sequence clean, add the trip, fly it again. A trip that works will collapse the variance term while leaving the mean largely unchanged.

7. What the aircraft can do about it

Detection is only useful if something can act on it. Three mitigations are available, operating on different timescales.

Passive: boundary-layer trip

A bead of tape on the upper surface forces transition upstream of the natural separation point. The resulting turbulent layer carries more near-wall momentum and stays attached through the adverse gradient. No bubble forms, so no bubble can burst.

Parameter	Value	Reasoning
Chordwise position	18–22% chord	Upstream of separation onset; sits on the rigid D-box, not the membrane
Height	0.5–0.8 mm	Boundary layer is 0.94–1.54 mm thick at that station across the cruise range
Spanwise extent	Full sheeted span	Re varies only modestly root to tip, so uniform sizing is adequate
Section	Round bead	More tolerant of the flow reversal at stroke reversal than a zigzag

The trade is that a turbulent boundary layer carries more skin friction drag and may separate marginally earlier at very high AoA. That cost lands on hover performance, where the margin is thinnest. This is exactly why it should be measured rather than assumed.

Adaptive: genome avoidance

Once H-LSB-1 is scoring, the genome learns which regions of the AoA–frequency–amplitude space are bubble-prone and weights away from them. This requires no hardware change and improves over the life of the aircraft, but it can only work within the envelope the pilot commands.

Structural: the membrane already helps

The compliant membrane aft of the D-box oscillates at small amplitude under unsteady load. That motion continually perturbs the boundary layer over the aft two-thirds of the chord, which disrupts bubble formation there. The membrane is already providing partial protection — but only downstream of the D-box. The rigid forward third, where the suction peak and the steepest gradient both live, has no such mechanism. That region is precisely where the trip tape would go.

8. Implementation scope

The addition is contained. Every piece of supporting infrastructure it requires already exists in the codebase.

Component	Status
One hypothesis entry in AERO_HYPOTHESES	New
Three SG task definitions	New
One ALSS segment, Class B curriculum	New
Cross-wing correlation computation in the tick	New — approximately 40 lines
Per-wing current stream at 50 Hz	Exists
Welford online variance (archaeology modality)	Exists — reusable
ALC tier-1 and tier-2 alerting	Exists
Genome routing and CMTS registration	Exists — pattern established
Persistence via persist_save_all	Exists

The correlation computation is the only novel logic. It maintains a rolling window of per-wing current residuals — actual minus the stroke-cycle-averaged expectation — and computes mean pairwise correlation across the eight channels at each evaluation. That value gates whether an elevated-variance sample is attributed to H-LSB-1 or to H-GS-1.

Recommended sequence

1. Fly the baseline. H-PERF-6 establishes the envelope; H-LSB-1 then gathers variance statistics on the clean wing across the cruise and glide envelope.

2. Add the trip tape and repeat the identical hypothesis sequence. The variance metric gives a direct before-and-after comparison, and H-PERF-3 reports what it cost in hover.

3. Let the genome converge. With bubble stability scored, the search naturally avoids the affected parameter regions, and the stability basin it finds becomes part of the persistent genome.

H-LSB-1 would be hypothesis 163. It closes the last open item from the adversarial physics audit — the one phenomenon known to be present in this Reynolds regime that the learning system had no framework to recognise.